

Setup & Startup

What the app is made of, and what happens in the first few seconds

⚠ FOR UNDERSTANDING PURPOSES ONLY

Prasanaa Kartick T — Personal Learning Project

Flutter / Dart

Packages	Theme	Splash screen	Database start-up
----------	-------	---------------	-------------------

1. What This Part Covers

Before any screen appears, a few behind-the-scenes pieces decide what the app is called, which ready-made tools it borrows, how it looks, and what the user sees first. None of them hold well data — they are the app’s “packing list, ID card and front door”.

Part	Plain-English role
Project settings	The packing list: name, version, borrowed tools, bundled pictures and fonts, launcher-icon recipe
App entry point	The “light switch”: the first thing that runs, plus the app-wide dark theme
Splash screen	A full-screen branded introduction image shown for about three seconds
Shared bottom bar	The Home / Operations / Safety bar reused across screens

2. The Packing List

The app is version **1.0.0** (build 1) and is marked private, so it can’t be accidentally published to the public Flutter package store. It borrows well-tested tools from other developers instead of building everything from scratch:

Tool (library)	What it gives the app
Local database	A built-in database on the device that stores equipment records, saved sheets and submitted reports
Desktop database support	Lets the same database work when the app is run on a Windows, Linux or Mac computer
PDF maker and printing	Builds field sheets, permits and reports as A4 PDFs and sends them to print or share
Excel maker	Creates spreadsheet exports

Tool (library)	What it gives the app
File sharing	Opens the phone's normal Share menu (email, WhatsApp, Drive...)
Folder finder	Finds safe places on the device to write generated files
Link launcher	Opens email and professional-profile links from the Contact screen
Advertising library	Included in the dependency list for ad support
Fonts and icons	A bundled Noto Sans typeface and an Apple-style icon set

Bundled pictures and fonts

The finished app carries its own artwork: the Drill Tick logo and splash image, contact icons, rigging and hitch diagrams used in the Rigging & Slinging module, well schematics for the kill sheets, equipment photographs for the inspection guides, and inspection-level illustrations (Levels 1 to 4). The Android and iPhone home-screen icon is generated from a single square image with a black adaptive background.

3. What Happens When the App Opens

1. The phone runs the app's single starting point, which first makes sure Flutter is ready.
2. If the app is running on a desktop computer, it switches the database engine to its desktop version. On a phone this step is skipped.
3. The app shell is created: the name "Drill Tick", the theme, and the Splash Screen as the first page.
4. The splash fades in over about 0.6 seconds, holds, then fades out over the final 0.6 seconds of its 3.2-second run.
5. Home then fades in over 0.4 seconds.

The theme

The app uses a **dark look** throughout: a deep navy background and a vivid green as the main accent. Individual modules add their own accent colours on top of this base, forming a colour language that is explained in the Home & Navigation document.

Nothing is read at start-up

The splash screen is purely visual. The database opens later, the first time a screen needs to save or load something, so the app starts quickly even if the database is large.

4. The Shared Bottom Bar

A slim black bar with three green icons — **Home**, **Operations** and **Safety**. Tapping Home returns to the first screen; tapping Operations or Safety clears the stack of open screens and opens that hub.

Because it returns to the root first, the back button never ends up in a confusing loop of previously opened screens.